

Nelson Chapter 6 Definitions

chemical kinetics the area of chemistry that deals with rates of reactions

reaction rate the change in concentration of a reactant or a product of a chemical reaction per unit time

average reaction rate the change in reactant or product concentration over a given time interval

instantaneous reaction rate the rate of a chemical reaction at a single point in time

catalyst a substance that alters the rate of a chemical reaction without itself being permanently changed

biological catalyst a catalyst made by a living system

heterogeneous catalyst a catalyst in a reaction in which the reactants and the catalyst are in different physical states

homogeneous catalyst a catalyst in a reaction in which the reactants and the catalyst are in the same physical state

collision theory the theory that chemical reactions can occur only if reactants collide with proper orientation and with enough kinetic energy to break reactant bonds and form product bonds

activation energy (E_a) the minimum energy that reactant molecules must possess for a reaction to be successful

activated complex an unstable arrangement of atoms containing partially formed and unformed bonds that represents the maximum potential energy point in the change; also called the transition state

rate law the mathematical expression that allows calculation of reaction rate as a function of reactant concentration

rate law equation the relationship among rate, the rate constant, the initial concentrations of reactants, and the orders of reaction with respect to the reactants

rate constant determined empirically and is unique for a single reaction at a specified temperature

order of reaction the exponent used to describe the relationship between the initial concentration of a particular reactant and the rate of the reaction

total order of reaction the sum of the exponents in the rate law equation

initial rate the rate determined just after the reaction begins (just after $t = 0$ s)

elementary step a step involving a one-, two-, or three-entity collision that cannot be explained by simpler reactions

reaction mechanism a series of elementary steps by which a chemical reaction occurs

rate-determining step the step in a reaction mechanism that determines the rate of the overall reaction; the slowest step in a reaction mechanism

reaction intermediate an entity that is neither a reactant nor a product but is formed and consumed during the reaction sequence